

Professional Summary

Enthusiastic and detail-oriented Data Engineering graduate student with hands-on experience in building real-time data pipelines, designing cloud-based solutions, and implementing scalable ETL systems. Proficient in Python, SQL, and AWS. Successfully executed 4+ end-to-end data projects including real-time streaming and ML pipeline deployments. Actively seeking full-time roles in data engineering or analytics.

Education

University of North Texas, Denton, TX

M.S. in Data Engineering, Aug 2023 – May 2025

Mallareddy College of Engineering, Hyderabad, India

B.E. in Electronics and Communication Engineering, Aug 2018 – Aug 2022

Technical Skills

Languages: Python, SQL

Big Data Tools: Hadoop, Spark, Apache Kafka

Databases: MySQL, PostgreSQL

Cloud Platforms: AWS (S3, Redshift), Google Cloud

Tools & Frameworks: Apache Airflow, Git, Jupyter, Tableau

Operating Systems: Linux

Projects

- **Real-Time E-Commerce Analytics Pipeline**

Tools: Apache Kafka, Spark Streaming, PostgreSQL, AWS S3, Apache Airflow

- Developed a real-time ETL pipeline to ingest and analyze clickstream data using Kafka and Spark Streaming.
- Reduced query latency by 40% through efficient data processing and PostgreSQL storage.

- **Customer Churn Prediction System**

Tools: Python, scikit-learn, pandas, NumPy, matplotlib, seaborn

- Built a machine learning pipeline to predict telecom customer churn with 85% accuracy.
- Performed data cleaning, feature engineering, and model selection using scikit-learn.

- **Smart IT Inventory Management System** [Project Link: <https://github.com/Jamescruz424/IT-Inventory-Management/blob/main/Zero-One-Sprint-2.pdf>]

Tools: Flask, React.js, Firebase Firestore, Python, JavaScript, GitHub

- Created a full-stack inventory tracking app using Flask (backend) and React.js (frontend).
- Integrated QR code and barcode support for efficient asset tracking.

- **Distributed IoT Analytics Platform**

Tools: Apache Kafka, Apache Cassandra, Python, Prometheus, Grafana

- Designed and implemented a real-time IoT data ingestion pipeline using Kafka.
- Deployed a scalable multi-node Cassandra cluster with TTL for sensor data.